# Vision Transformer-Based Visual Analysis of Hybrid-Slotted DGS Compact Patch Antennas

Module 1 of the Applied Vision Transformer Framework for 5G and Emerging 6G Networks - Proof-of-Concept Prototype

*[Student name] | [Student ID] | [Programme / Module] | [Institution] | Applied AI Research Workshop - Module 1, October 2026*

## 1 Research objective

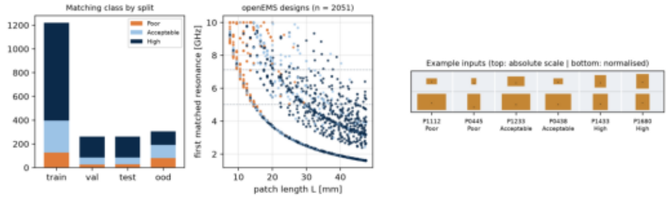
- Can standardised antenna geometry images predict simulated performance classes - and show where the model looks?
- Module 1 = image branch of the two-module framework; Module 2 (surrogate, fusion, NSGA-II) is a separate study.
- Proof of concept on public simulation data; Ansys HFSS was not available and was not run.
- Hypotheses H1-H5 committed to Git before training.

## 2 Data & preprocessing

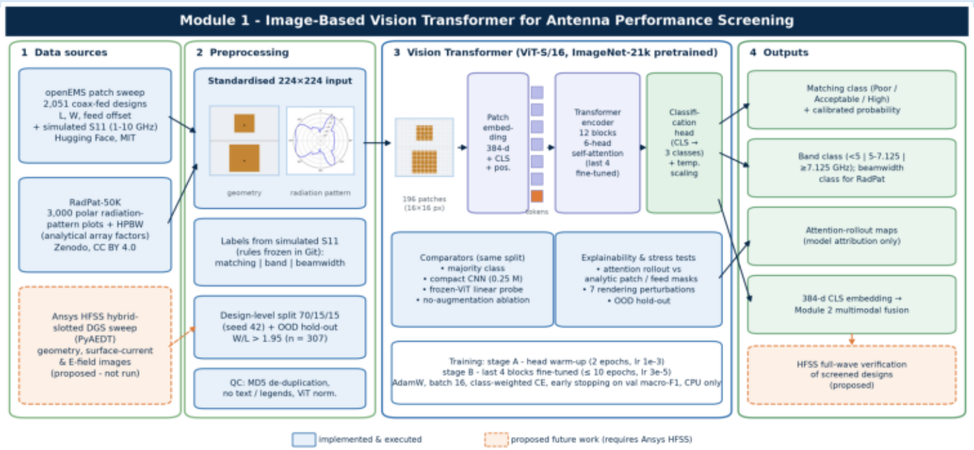
- openEMS patch sweep (Hugging Face, MIT): 2,051 designs -> one 224x224 two-panel image each
- RadPat-50K (Zenodo, CC BY 4.0): 3,000 polar radiation-pattern plots -> beamwidth class
- Locked design-level split 70/15/15 (seed 42): 1,220 / 262 / 262 + OOD hold-out W/L > 1.95 (n = 307)
- No text / legends, MD5 de-duplication, training-only mirror, ±3° and 2% shift

**Matching class: Poor > -10 dB | Acceptable -20 to -10 dB | High <= -20 dB***

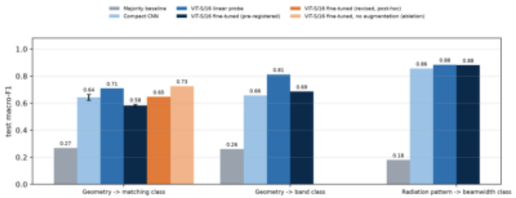
**-20 dB boundary is a project assumption awaiting engineering confirmation*

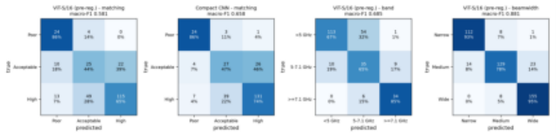



## 3 Module 1 block diagram (implemented vs proposed)




## 4 Results on the locked test split (executed)

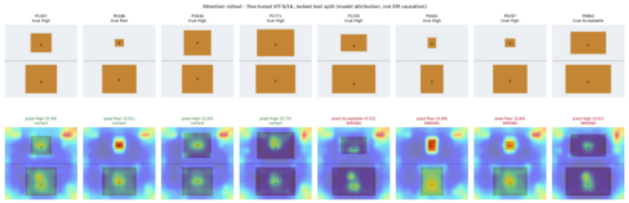



| Model (geometry -> matching class) | Macro-F1 | Bal. acc. | Other tasks |
|---|---|---|---|
| Majority baseline | 0.269 | 0.333 | |
| Compact CNN (3 seeds) | 0.643 ± 0.022 | 0.669 | Band class |
| ViT-S/16 linear probe | 0.709 | 0.729 | ViT 0.685 |
| **ViT-S/16 fine-tuned (3 seeds)** | **0.582 ± 0.008** | **0.649** | probe 0.811 |
| ViT fine-tuned - OOD hold-out | 0.679 ± 0.027 | 0.664 | RadPat beamwidth |
| ViT, no augmentation (ablation*) | 0.725 | 0.737 | ViT 0.881 |

**exploratory, 1 seed, not pre-registered; underfitting traced to augmentation blurring the 1-2 px feed offset*

## 5 Explainability (attention rollout)



Median enrichment vs area share: feed halo ×1.41, patch ×1.03

*Attribution only - not proof of electromagnetic causation*

## 6 Robustness & key insight

- Blur (1 px) -> 0.526, grey copper -> 0.521 (clean 0.581); contrast and JPEG barely matter
- OOD 0.679 > test: class balance differs, so the shift is confounded - not better generalisation
- Without augmentation the same ViT reaches 0.725: augmentations must preserve fine geometry

## 7 Hypotheses & conclusion

| | | |
|---|---|---|
| **H1** | ViT > majority baseline (both patch tasks) | **Supported** |
| **H2** | ViT ≥ compact CNN (3-seed mean, matching class) | **Not supported** |
| **H3** | Partial fine-tune > frozen linear probe | **Not supported** |
| **H4** | Attention enriched on patch and feed (median > 1) | **Supported** |
| **H5** | Lower macro-F1 on OOD (W/L > 1.95) | **Not supported** |

- Working, reproducible image pipeline - ready for HFSS data
- Next: >= 1,000 HFSS hybrid-slotted DGS designs (PyAEDT)

**Explainable image-based screening of compact patch antennas for 5G / 6G - simulation evidence only, not HFSS-validated**

Data: LaBash et al. (2025) arXiv:2505.18188 [MIT] | Bhuma et al. (2025) RadPat-50K, doi:10.5281/zenodo.17067881 [CC BY 4.0]
ViT: Dosovitskiy et al. (2021) | Attention rollout: Abnar & Zuidema (2020) | Calibration: Guo et al. (2017) | Code: https://github.com/rathiabeth-sudo/901---module-1